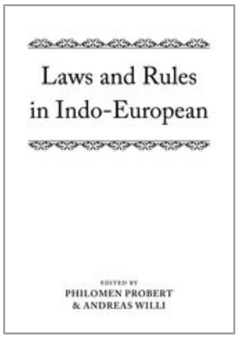

Laws and Rules in Indo-European
Philomen Probert and Andreas Willi

Print publication date: 2012
Print ISBN-13: 9780199609925
Published to Oxford Scholarship Online: September 2012
DOI: 10.1093/acprof:oso/9780199609925.001.0001

Older Runic evidence for North-West Germanic a-umlaut of u (and ‘the converse of Polivanov's Law’)

Patrick Stiles

DOI:10.1093/acprof:oso/9780199609925.003.0004

Abstract and Keywords

The main structural effect of North-West-Germanic ‘a-umlaut’ — which involved shuffling the allophones of /i/, /e/ and /u/ depending on the following environment — was creation of the phoneme short /o/, split from Proto-Germanic /u/, as in Old High German gold “gold” beside guldīn “golden”. The standard view sees the change as an instance of secondary split, phonologized by the loss of one of the conditioning factors, short a, from final syllables, thus: nom. sg. */nistaz/ [nestaz] > /nest(z)/ “nest”. Up to then, the allophones were in complementary distribution. However, such forms in the Older Runic language (c.150 to 500 AD) as horna (Gallehus), worahto [= worahto] (Tune); holtijaR (Gallehus), dohtriR (Tune) call this account into question by showing the change with the supposed conditioning factors intact. The chapter seeks to explain this state of affairs, which is in apparent

conflict with 'Polivanov's Law', which states there is 'no split without a merger, that is sound-changes that add to the inventory of phonemes are always triggered by a phonemic merger (including with zero, i.e. loss) elsewhere in the system.

Keywords: Old Germanic languages, sound change, a-umlaut, phonologization, Older Runic, Prague School linguistics, PHONEMIC split, Polivanov's Law

4.1 Details of the change and its main structural effect

4.1.1 Introductory

North-West Germanic *a*-umlaut has been much discussed, yet it seems to me that some of the most interesting and fundamental problems have not received due attention. These are posed by the evidence of Older Runic inscriptions.

North-West Germanic is an intermediate proto-language, ancestor of both the North Germanic (Scandinavian) and West Germanic languages, which are attested in manuscripts from about 1150 and shortly before 700 AD, respectively, and which both have earlier runic traditions. They constitute two of the three branches of Germanic, the third being East Germanic, in effect Bishop Wulfila's Bible translation from the fourth century AD, which is attested in manuscripts written two centuries later. Older Runic is used to designate the (early North Germanic) language of runic inscriptions written in the Older Futhark alphabet, from the earliest examples, *c*. (1)50, up to *c*. 500 AD.[1]

(p.44) This article is not intended as a contribution to linguistic theory, but to understanding how *a*-umlaut might have been phonologized. I lay no claims to originality in my analysis. Most of what I shall present lies buried in the secondary literature in the context of other topics, and I have done little more than disinter it. The ideas are hardly revolutionary; they have just not been applied to the specific problem of *a*-umlaut and are not reflected in our handbooks of Germanic. I have kept references to a minimum and concentrated on earlier work (cf. n. 3).

4.1.1.1 Mechanism

The term '*a*-umlaut' is used here—faute de mieux—as an inaccurate but convenient label to refer to a set of early assimilatory changes in North-West Germanic affecting the high and mid short vowels in accented syllables.[2] The Proto-Germanic inventory had four members: /i/, /e/, /a/, /u/ (cf. Benediktsson 1967). The change involved the lowering and raising of short *i* and *e* and the lowering of short *u*, depending on the one hand whether a high or low vowel followed in the next syllable and on the other whether a nasal-initial consonant cluster immediately followed. (The vowel *a* was unaffected by these changes.) The diphthong /eu/ was also subject to *a*-umlaut, but will not be of concern here.

4.1.1.2 Environments

The relevant environments are three in number: those that conditioned a high vowel, a low vowel, or were neutral. They are constituted as follows:

(a) High vowel: before *i*,*ī*, (*iu*) and *j* in the following syllable, and when the intervening consonantism starts with a nasal, regardless of the vocalism of the following syllable. (The sequence *-*eww*- also became *-*iggw*- (p.45) regardless of the following vowel: see Heidermanns 1986: 297–9, Rasmussen 1989*a*; cf. Kock 1910: 98. Similarly, only the sequence -*ugg*(*w*)- occurs; cf. Kock 1898: 517.)
Examples: **giƀiđ* 'gives' (beside *geƀan* 'to give'), **hringaz* 'ring' (〈 **hrengaz*, cf. Finnish *rengas*), **hundaz* 'dog'.
(b) Low vowel: before short *a*, long *ō* (and long *ā*, in so far as it existed). (Proto-Germanic absolute final bimoric long *-*ō*# had already shortened to *-*u* in North-West Germanic by the time of *a*-umlaut.)
Examples: **weraz* 'man' (〈 **wiraz*, cf. Lat. *vir*), **snozō*- 'daughter in law' (〈 **snuzō*-), **golþa* 'gold' (beside **gulþīn*- 'golden'), **korna* 'corn' (〈 **kurna*), **worhtō* 'I worked, made' (beside **wurhtē* 3sg.).

(c) Neutral: before *u*,*ū*, and *ē* (also short *e*, if it existed in unaccented syllables). Also when nothing followed (in monosyllables).
Examples: **siđuz* 'custom', **međuz* 'mead', **wig* ipv. 'kill!', **weg* ipv. 'carry!', **sehs* 'six', **snuzu* nom. sg. 'daughter in law',**duhtēr* nom. sg. 'daughter'.

As can be seen, in environment (c) the vowel was unaffected and the etymological value was retained. This means that although /i/ and /e/ were to a large extent in complementary distribution (cf. 4.1.1.3), they did not fully merge as phonemes (cf. for example Beeler 1966, Benediktsson 1967: 190–4; already Luick 1914–40: 1.i. §71 A2).

4.1.1.3 Distribution of allophones

The distribution of allophones of the four short accented vowels in the various environments can be summarized as in Table 4.1.

Table 4.1 Distribution of allophones of the four short accented vowels in North-West Germanic

	High vowel	Low vowel	Neutral
/i/	[i]	[e]	[i]
/e/	[i]	[e]	[e]
/u/	[u]	[o]	[u]
/a/	[a]	[a]	[a]

This schema implies a stage in the development of the North-West Germanic languages when there were automatic positional allophones [i] and [e] of /i/ and /e/, and [u] and [o] of /u/ in the raising and lowering environments just specified. (By contrast, *a* occurred in all environments.) During this 'phonetic period', (p.46) there were sub-phonemic variations within a paradigm, for example nom. sg. [duhte:r], but gen. pl. [dohtro:].

4.1.1.4 Creation of short /o/

The main structural effect of *a*-umlaut is the creation of the phoneme short /o/, split from Proto-Germanic /u/. (As concerns /i/ and /e/, the upshot is chiefly changes in the incidence of pre-existent phonemes.) Exploring the means whereby the single phoneme PGmc short /u/ came to be replaced by two phonemes, North and West Germanic /u/ and /o/, is the concern of this article.

4.2 The structuralist account and its inadequacies

4.2.1 The change as 'secondary split'

As the development of /u/ and /o/ from PGmc short */u/ entailed adding a new term to the phonemic inventory, the change is classified by structuralist theory as an example of secondary split (as opposed to primary split, where the 'splitting' allophones join pre-existing phonemes and the overall inventory is unaffected).[3] It is an axiom, known as Polivanov's Law,[4] that in a phonological system, there is 'no split without a merger'. That is to say: any split always involves a simultaneous merger of some kind. In primary split, the splitting-off allophone is the one that does the merging (and the process can also be styled 'partial merger'). In the case of secondary split, the merger takes place elsewhere, as the creation of the new phoneme is triggered by merger (including with zero) among the conditioning factors; loss of contrast there transfers the distinctive function to the conditioned phones.

4.2.2 The classic interpretation

The classic structuralist interpretation is that *a*-umlaut was phonologized by the loss (merger with zero) of one of the conditioning factors, short *a*, from final (p.47) syllables: thus, for example, nom. sg. /hurna/ [horna] 〉 /horn/ 'horn'. That is, the allophones were in complementary distribution until they were phonemicized by the loss of final -*a* (generally regarded as the vowel that was lost earliest from final syllables).[5]

This view is epitomized by Moulton (1961*a*: 14–15):

> Die komplementäre Verteilung von urgerm. *[i] und *[e], *[u] und *[o], *[iu] und *[eo] war von dem Weiterbestehen der unbetonten Vokale der folgenden

> Silben (*i, *u*, bzw. **a*) völlig abhängig. Solange diese unbetonten Vokale fortlebten, wurde die komplementäre Verteilung der Allophone aufrechterhalten. Sobald die unbetonten Vokale aber zusammenfielen (welches jedenfalls theoretisch hätte vorkommen können) oder schwanden (welches tatsächlich eintrat), war die komplementäre Verteilung der betonten Vokale durchbrochen.

> …um die phonetischen Oppositionen *[i e], *[u o], *[iu eo] zu phonologisieren, genügte es, daß die komplementäre Verteilung der Allophone in je einer Lautumgebung aufgehoben wurde. Der Verlust dieser unbetonten Vokale und damit die Phonologisierung der Allophone der betonten Vokale ist eine Erscheinung, die allen nord- und westgerm. Sprachen gemeinsam war.

(Moulton erroneously assumes complementary distribution; cf. 4.1.1.2–3.)

4.2.3 Older Runic counter-evidence

This account would be exemplary and unobjectionable, were it not for the existence of certain forms in the Older Runic language.[6] I cite as examples forms for which both reading and interpretation are fairly certain: **horna** 'horn' acc. sg. (Gallehus), **worahto** 'I made' [= *wor^{a}hto*] (Tune), **holtijaR** 'of Holt' (Gallehus), **dohtriR** 'daughters' nom. pl. (Tune).[7]

The first two examples have 〈**o**〉 (presumably *o*) correctly conditioned; the second two examples have 〈**o**〉 (presumably *o*) analogically transferred from a 'forme de fondation' to a 'forme fondée' and present an environment that 'ought' to have 〈**u**〉 (*u*).

(p.48) These forms confront the classic structuralist view with two problems:

> (1) Short *-a* has clearly not been lost—indeed its retention after heavy and light syllables both medially and finally is one of the defining characteristics of the Older Runic linguistic stage, cf. Grønvik (1998: 15)—so we would not expect the allophones to be graphically

> represented. We should have only 〈**u**〉 -spellings in all these instances, representing /u/, and the rune 〈**o**〉 should denote solely the long vowel /o:/.[8] The fact that the allophones are graphically represented indicates that the sound change of *a*-umlaut, paradoxically and ‘prematurely’, has somehow been phonologized *before* the loss of its conditioning factors. In addition, and in apparent contradiction of Polivanov's Law (cf. 4.2.1), we appear to have an instance of split without merger. If it is not the loss of the conditioning factors, what has phonemicized the allophone [o] of the phoneme */u/?
> (2) The distribution of allophones is patently *not* complementary, which again indicates that the sound change has been phonologized.

As scholars almost invariably remark, the forms **holtijaR** , nom. sg. of a derived adjective, and **dohtriR** , nom. pl., presuppose base-forms /holta-/* and /dohtar-/*.[9] But, although this statement is evidently correct, it cannot be said to be particularly illuminating for our purposes; the nub of the matter is how we are to account for such forms as /holta/* and /dohte:r/*. These forms, and /horna/ and /worhto:/, constitute the core of the problem that Older Runic poses for the classic structuralist view of *a*-umlaut; **holtijaR** and **dohtriR** are merely an epiphenomenon.

It is admittedly a very small corpus of four items—extendable to six if the base-forms just discussed are included—but adequate in the circumstances, I believe. That is, having more data would not necessarily tell us much more about the language. The two inscriptions come from south-east Norway (Tune) and south Jutland in Denmark (Gallehus), which involves a reasonable geographic spread.

4.2.4 Distribution of reflexes

There has been quite some discussion of the second problem mentioned in 4.2.3, that of the distribution of the reflexes, largely in the context of the question of ‘*i* and *e*, two phonemes or one?’ (a non-issue as they were clearly two, cf. 4.1.1.2–3), and the problem of ‘failure of *a*-umlaut of *i*’.[10] However, the focus here is on the phonologization of *o*.

(p.49) Of the West Germanic languages, High German is usually considered to have carried through the change of *u* to *o* most consistently, cf. OHG *wolf*, OE *wulf* ‘wolf’. Yet, even here, by-forms—e.g. OHG *snora, snur(a)*, Germ. *Schnur* ‘daughter in law’, etc.—indicate that later repartitions are the result of analogy. In some instances, where there were both semantic and phonological variants, the products could be lexicalized, cf. OHG *stehhal(a)* ‘drinking vessel’ and *stihhil(a)* ‘prick, sting (Germ. *Stachel*)’, cognate with Goth. *stikls* ‘drinking vessel’.

It should be noted that in North Germanic reflexes of the lowering of *u* to *o* show a dialectal distribution. There is a cline from west to east, with *o*-reflexes much more established in the west—in Icelandic, Faroese, Norwegian, and Danish west of the Great Belt (Jutlandic and on Funen)—than in Danish east of the Great Belt, Scanian (the dialect of Skåne), Swedish, and Old Gutnish (the language of the island of Gotland), where *u*-reflexes predominate. See Bandle (1973: 24–8), with important earlier references; compare also the discussion in Benediktsson (1967: 184–6). It is clear that the repartition is largely the result of secondary developments (cf. Kock 1910: esp. 138–9). Thus, even in Old Gutnish, which has one of the highest proportions of *u*-outcomes, the word for ‘daughter’ has preserved a reflex of **o* in certain forms of the paradigm, for example the accusative singular: *dōtur* (〈 **dohtaru*); while elsewhere showing a reflex of *u* levelled from case-forms providing a high-vowel environment: dat. pl. *dȳtrum* (see 4.1.1.2–3). This is no doubt because the *-*h*- was lost and the preceding vowel lengthened, fossilizing the alternation.[11]

Because of the nature of the evidence, the remainder of this article relates to Older Runic rather than the rest of North-West Germanic. However, the discussion is mainly about interpreting the evidence rather than the evidence itself.

4.2.5 The problem of phonologization

Relatively little attention seems to have been paid to the first problem mentioned in 4.2.3—how /o/ came to be phonemicized—except by some Scandinavian scholars whose response is to deny that there was any such thing as *a*-umlaut in the sense used here. (Part of the difficulty stems from a literalistic interpretation of *a*-umlaut to mean lowering before a following short *a* and nothing else, cf. n. 2.) These scholars operate instead with a lowering before certain consonants.

(p.50) Thus Harry Andersen (1960: 406) speaks of '*o*-forms such as **holtijaR** (which presupposes **holta* not **hulta*) **worahto** and **dohtriR** , which can have nothing whatsoever to do with a presumed *a*-umlaut' and concludes that the 'shift of *u* to *o* must be extremely old in the Germanic languages (compare Gothic) and it must have been independent of a following *a*'.[12] He and others posit a Proto-Germanic lowering of *u* to *o* before certain consonants, notably *r, h*—as in Gothic—(and to a lesser extent *l*).[13]

This claim does not stand up to scrutiny. In the first place, it is directly refuted by such West Germanic forms as OE, OHG *burg* '(fortified) town', OHG *suht* 'illness', *zuht* 'nourishment, breeding, offspring'; as well as Old Gutnish *dȳtrum* dat. pl. 'daughter', cited in 4.2.4.

Secondly, the fact that Proto-Germanic had no /o/-phoneme means that if there was a lowering of /u/ before certain consonants, it must have been sub-phonemic. Yet, proponents of the lowering view offer no account of how a short [o]-allophone was phonemicized. (I would not deny the possibility that a lowering influence of *h*—and *r* and *l*—may have worked in such a way as to favour the generalization of *o*-variants, once the phoneme /o/ existed. Consonantal environments apparently played a role in determining lexicalized forms of words; cf. Bandle 1973: 25–7, with references.)

These scholars are right to have raised objections, for, as we have seen, there *are* problems the 'standard view' ignores. Grønvik (1998: 86) notes that 'diese Allophone [wurden] phonemisiert, ohne daß dieser Vorgang bisher völlig geklärt wäre. Diese Phonemisierung ist jedoch schon für die älteste Runensprache…sicher nachzuweisen.' The problem should at least be addressed, even if it cannot be satisfactorily answered.

4.3 Ways in which allophones achieve phonemic status

4.3.1 The standard view

Conveniently for the present purposes, in a couple of articles Dobson explicitly considered the question of how it is that allophones come to be phonemicized (1962: 141–4, essentially repeated in 1969: 43–8). Two of the ways he gives in which 'a contextual variant, an allophone, comes to be regarded as a distinct (p.51) sound in its own right, a phoneme separate from the one from which it arose' (1962: 141) are of relevance to changes that add a phoneme to the inventory (cf. 4.2.1). These are (cf. 1969: 43):

> (1) destruction, by some independent sound change, of the environment in which the allophone arose, so that it ceases to be tied to an existing phonetic context
> (2) an allophone of one phoneme developed in one context or set of contexts may become phonetically identical with an allophone of a different phoneme developed in a different context or set of contexts, in such a way that the merged allophones are neither referable to a single source nor identifiable as occurring in limited and recognizable contexts.

The first method is prototypical secondary split; the second could result in merger with an existing third phoneme (and thus be a kind of 'partial merger') or it could lead to creation of a new phoneme (for example, in a given language, rounding of /e/ and fronting of /o/ could coincide and produce a previously non-existent /ö/; in those circumstances, the split would also be the product of a merger, although not among the conditioning factors). What is clear, however, is that neither of these routes can account for the phonologization of *a*-umlaut (cf. 4.2.3).

Dobson goes so far as to exclude a further way in which an allophone could become a phoneme (perhaps because he conceives of this as a general process that comes about 'when the variant ceases to be recognizably dependent on the context or contexts in which it originally developed', 1962: 141). In his later article (1969: 44), he writes:

> analogical processes, as when a variant developed in one inflexional form is transferred to another, must always take place after a distinction has become phonemic;[3] the possibility of transference depends on observation and choice, and the lack of an automatic link with the context. Thus in primitive OE there must already have been a phonemic distinction between *æ* and *a* before **færiþ* could be remodelled as **fariþ* (to become *færþ*); it is the business of the phonologist to ask by what means it had become phonemic, for it certainly does not depend on the mere physical difference between the sounds, which is slight.
>
> [3]Explanations which assume the analogical substitution of one allophone for another are contrary to principle…

(Cf. Dobson 1962: 143–4 n. 1.) Trnka ([1936 =] 1982: 30) seems to have been the first expressly to deny the possibility of analogical transference of allophones (also with reference to the Old English *a* ≠ *æ* opposition).

4.3.2 Phonemicization by levelling

However, Bazell, for one, had already dissented from this position, claiming that an allophone can indeed achieve phonemic status by ‘levelling’ (1952: 37):

> (p.52) The analogical transference of an allophone is by definition impossible, since the ‘allophone’ would then have phonemic status. Occasionally (although this is not a very productive source of new phonemes, and is even denied entirely by some scholars) new phonemic distinctions arise in this way. Thus the Old English short vowels *a* (before back vowels of the following syllable, including the combination *-ia-*) and *æ* (elsewhere) appear to have been variants before analogical forms (*fare* subj. pres. after infin. *faran* etc., as against *fære* dat. sing. of *fær*) arose.

Cf. also the discussion in Hoenigswald (1944: 82–3).

Bazell, I think, must be correct (probably also specifically for the *a* ≠ *æ* opposition in West Saxon Old English; similarly

Samuels 1972: 35). Yet this observation still begs the question of how it can be that speakers should start shuffling around a supposedly automatic alternation.

4.3.3 Alternations can cease to be automatic

On this topic, C. A. Ladd (1964: 656) offers an interesting insight that adds detail and clarification to Bazell's statement. In brief, the alternation must cease to be wholly automatic—if by 'automatic' we mean that speakers are unaware of it. After a lapse of time, the allophonic rule that gave rise to the alternation is no longer 'live' and

> a new generation of speakers hears the allophones as two sounds rather than one. We cannot suppose that children have to wait for a minimal pair before they decide whether two sounds form two separate phonemes or not; what matters is that they can interchange them freely…children learning the language…imitate the sounds in the distribution in which they heard them, a distribution which depends on phonetic criteria. At the same time, however, the way is now open for analogical interchange which will disturb the phonetic pattern. It is in fact the possibility of disturbance which indicates that the change has reached the phonemic level; the two things go hand in hand. Frequently, it is the appearance of analogical forms which leads to the recognition that the sound-change has occurred…

4.3.4 The role of 'system balance'

Some remarks of Kuryłowicz regarding the phonologization of '*a*-umlaut' are of relevance to how speakers might have become aware of the previously automatic alternation in this instance (1952*b*: 54):[14]

> (p.53) The rise of long *ā*…from *ē* $_1$…severs the quantitative bond between [short] /a/ and [long] /o:/, the latter becoming the long grade of [o]. The vocalic timbres [o] and [u], up to now variants of a single phoneme /u/, become INDIRECTLY autonomous owing to the phonological distinction between the respective long vowels /o:/ and /u:/…

Kuryłowicz incisively draws attention to the relevance of the North-West Germanic rise of a new long /a:/ phoneme (by shift). This had the effect of changing the correlations between short and long members of the phonological space. Linking the development of the short vowels to the shift of long /e:/ to /a:/ (except in final syllables) certainly correlates with the fact that *a*-umlaut is a non-Gothic North-West Germanic feature (cf. n. 2).

Elmer H. Antonsen developed Kuryłowicz's thought and interpreted it as demonstrating how [u] and [o] were phonemicized by what Antonsen came to call 'system balance' (e.g. 1965: 27, 1972: 138, 2002: 25). In his 1965 article, he noted that /o/ filled the 'hole in the pattern' of short vowels (compare also Benediktsson 1967: 177, 1970: 102). Introduction of a phoneme as a result of the re-evaluation of allophones in terms of the oppositions in the phonological system has been discussed several times by Moulton (his study 1961*b* offers an interesting analogue to the case of *a*-umlaut).[15]

Awareness of the distinction between the allophones is the necessary precondition, but levelling (which may itself have been 'unconscious') was the decisive step that gave phonological status to the variants by disrupting the complementary distribution.

4.4 Diachronic considerations and the question of reversion

4.4.1 The transmission problem

The factors outlined in 4.3.2–4 help explain how *a*-umlaut came to be phonologized before its conditioning factors were lost, in defiance of Polivanov's Law. Dobson—and, of course, many others; I am only using him as the representative of a viewpoint—had a too rigidly synchronic concept of the phoneme, and (as a corollary) an almost a-chronistic approach to its application to historical phonology (cf. 4.3.1). Language may be a system 'où tout se tient', as de Saussure maintained (see Koerner 1999), but it is rather a loose system and 'all grammars (p.54) leak' (Sapir 1921: 38), even synchronic ones. If this were not so and they were 'watertight', there would be no redundancy and the loss of a phoneme, indeed any real change, would be impossible (cf. Hockett 1965: 203). The reality is continuous change and, even in a synchronic analysis, we are trying to hit a moving target.

The simplistic transfer of methods of synchronic analysis on to the diachronic plane is reminiscent of Zeno's paradoxes (cf. Coseriu 1958: 10–12 = 1974: 12–14). One of these is taken up memorably by Tolstoj, in the opening section of *War and Peace*, Book Three, Part Three: operating with discrete units of motion instead of continuous motion has the result that —'logically'—Achilles can never overtake the tortoise, who has been given a start, although we all know in reality that he can. Likewise—puristically—the *i*-mutation of short *o* is 'impossible', because, according to the rules of *a*-umlaut (cf. 4.1.1.2), only *u* could stand before an *i*-mutation factor. Yet, of course, the *i*-mutation of *o* is well attested, being reflected, for example, in OIc. *dǿtr* 'daughters' (nom. pl.). Janda and others have fallen into the same 'logical' trap (cf. Joseph and Janda 2003: 409–10 with references) in assuming that the loss of conditioning factors 'ought' in every case to result in reversion of allophones; cf. 4.4.2–3. This is a diachrony that is strangely devoid of history.

Even granting the correctness of the dogma that all speakers are totally unaware of subphonemic alternations, this could only really be so synchronically; it ignores the transmission of language from generation to generation (as noted by Ladd, cf. 4.3.3)—recognizing, of course, that generations form a continuum within the community, cf. Weinreich et al. (1968: 114) (already Saussure, *Cours de linguistique générale* I.2. §1 = 1955: 106).

4.4.2 'Significant allophones'

I think a point C. A. Ladd makes about Old English palatalization earlier in the article cited previously is a crucial one (1964: 655):

> what happens when a subphonemic variation becomes phonemic…is a vital stage in the process which has received curiously little attention…In the case of

> conditioned sound-change the result will eventually be a phonemic split of some sort. Now, it is surely not sufficient to say that in such cases a subphonemic variation becomes phonemic as the result, for instance, of a loss of conditioning sounds. In the case of the split between back and front *c* and *g* in Old English there is little question that other consonants than *c* and *g* originally had fronted allophones, as the same sounds which produced fronting of *c* and *g* also produced front mutation of the vowel of the preceding syllable, no matter what the intervening consonant. But only in the case of *c* and *g* had the variation between the two allophones reached such a degree that they could stand on (p.55) their own without the support of the conditioning sounds. In other words, even before the loss of the conditioning sounds, the two allophones must have been potentially phonemic…

It is probable that these reconstructable 'significant allophones', which hindsight reveals to have been 'potential phonemes', correspond synchronically to what may usefully be termed 'perceived allophones'.[16] Instead of an absolute divide between the phonemic and subphonemic, there is rather a cline from unconscious 'universal phonetic' accommodations, through unnoticed automatic alternations to perceived allophones (the dividing line between these two probably varying somewhat from speaker to speaker), and on to actual phonemic contrasts (the acid test being that they are actually exploited functionally). Here, there is also a range, from 'marginal phonemes', with minimal, to 'full phonemes', with maximal functional load.[17]

As C. A. Ladd has pointed out, minute changes in pronunciation cannot spread consistently through a speech community 'unless they spread by (p.56) imitation'; further, 'to be imitable, variations in pronunciation must be perceptible' (1965: 100–1; cf. also Weinreich, Labov, and Herzog 1968: 130–1). If this were not so, mimicry (the imitation of someone else's speech idiosyncracies), let alone code-switching, would be impossible.

4.4.3 Allophone formation and reversion

Reversion, a subject that Ladd raises in the quotation given in 4.4.2, is a genuine issue. In many cases where split would be possible, the presumed allophonic variation effectively vanishes and no phonemicization takes place when the ‘conditioning factors’ are lost (cf. Hoenigswald 1999: 204). Polivanov had already observed that ‘split can only take place when there is sufficient physical difference between the diverging variants’ (1928: 38 = 1974: 77). Similarly, Hoenigswald added a caveat concerning secondary split: ‘allophones become phonemes when part or all of their determining environments fall together without at the same time canceling the phonetic difference between the allophones in question’ (1960: 94, also 77 ‘if the phones remain distinct’; cf. 1999: 204). Compare also the discussion of Janda (in Joseph and Janda 2003: 409–10, with references). Van Wijk, in the early days of Prague structuralism, explicitly recognized that whether phonologization takes place in a given instance is largely a matter of phonetics and thus subject to chance (e.g. 1939: 304–5).[18] Much the same point has been made by Ohala (e.g. in Joseph and Janda 2003: 683–5; perhaps it is not surprising that those who consider these issues reach similar conclusions). This underlines the fact that our knowledge is imperfect and little more is open to us than post-hoc rationalization.

Another factor to bear in mind when considering the apparent reversion of allophones that one might have expected to be phonemicized is the process of allophone formation. A prime example, which is scarcely mentioned by the handbooks, concerns the phonologization of *i*-umlaut in Old High German —or rather the non-phonologization of *i*-umlaut at the time of the loss of short -*i* after heavy syllables. This is certainly different from the situation in Old English: compare OE *giest, giestas* ‘guest, guests’ and *mann, menn* ‘man, men’, beside OHG *gast, gesti* and *man, man*. In view of the outcome of ‘primary umlaut’ (as in *gesti*), it is hard to see why supposed umlaut allophones (of *a*, at least) should have reverted before lost -*i*. Rather than positing wholesale reversion—shortly to be followed by phonemicization of other apparently ‘identical’ (p.57) conditioned variants—it seems more plausible to suppose that 'significant' *i*-umlaut allophones had not yet formed when short *-i* was lost after heavy syllables, but developed later, and were phonemicized by the loss of **-(i)j-* (including *-ja-* 〉 *-e-*), which causes *i*-umlaut in both English and German, e.g. *to drench : tränken; to hear : hören*. (On the subject of allophone formation, we may further note that Russian shows no sign of developing an allophone [ŋ] in words such as *bank* 'bank': [bank], not †[baŋk].)

4.4.4 Citation forms and the spoken chain

It is also worth recalling that sound change originates not in citation forms but in the spoken chain. The allegro forms of connected speech are characterized by special allophones conditioned by their environment in the utterance, and varying degrees of reduction (utterances are often distinguished by distinctive features that do not play any part in the composition of the 'paradigmatic' phoneme inventory). In 'decoding speech', the listener evaluates them against the template of the lento citation forms (which conventionally serve as the basis of phonemic analysis).

At certain stages, when the divergence gets too big for 'comfort' (ease of comprehension), some sort of accommodation takes place in (sections of) the speech community. Either (somewhat more rarely, I should guess), speakers adjust their 'contextual forms' so as to bring them more into line with the citation form ('don't speak in such a slovenly fashion, dear'), or (the most usual strategy), the citation form is restructured to bring it more into line with the contextual form—a fact to which Polivanov already drew attention (1928: 39–40 = 1974: 77–8; cf. also Hockett 1965: 203). This must be what happens in the case of such 'radical' changes as *i*-umlaut, when a whole new class of (front rounded) phonemes enters the language and, for a period, the new forms that contain them compete with the 'original' forms.

This is one way in which sound change can take place within purely chronological 'space' (cf. C. A. Ladd 1964: 656, quoted in 4.3.3), even where there is little relevant geographical or sociological variation in a given speech community (e.g. a remote island containing a single village of egalitarians). Greater geographical dispersal and social stratification of a single speech community will certainly lead to widespread 'dialect borrowing' and a greatly increased rate of change, which again can be somewhat offset by literacy, standard languages, and so forth.

I have been concerned with some aspects of the 'generation of innovations' rather than the 'propagation of change' in all its sociolinguistic aspects, something we cannot reach in reconstructive work.

(p.58) 4.5 Main conclusions

4.5.1 Review of how a phoneme can be introduced into the inventory

To round off the discussion, I rehearse the various ways in which a new phoneme can be introduced into the inventory:

I. As the result of 'regular sound change', which is to say by secondary split, to which Polivanov's Law applies (cf. 4.2.1, 4.2.3(1)).
II. As the result of some kind of 'analogy' ('split without merger'), where—because it is not strictly a sound-change process—Polivanov's Law does not apply:

(a) The redistribution of perceived allophones by levelling within a paradigm and/or between a base and a (synchronic) derivative. This is the process I have invoked for NWGmc *a*-umlaut, where the allophones had come to be perceived through 'system balance' (4.3.4). Levelling of this kind is often tacitly adduced in historical work, but it is unclear to what extent it is explicitly acknowledged in historical linguists' theory of what they do. As in conventional secondary split (I), the variants must have been phonetically conditioned and phonemicization results from disruption of complementary distribution. This does not hold for the remaining two analogical types.

(b) 'Morphophonemic analogy'—the redistribution of allophones by pressure of an existing morphophonological alternation. Such a development was first suggested by Jakobson (1949: 17 = 1971: 114), with the example of Byelorussian palatalized consonants. The existence of the alternations /rv-ú/ 'I tear' /rv'-oš/ 'you tear', /vr-ú/ 'I tell lies' /vr'-oš/ 'you tell lies' calls into being the pair /tk-ú/ 'I weave' /tk'-oš/ 'you weave' (replacing /tk-oš/) and creates the phoneme palatalized /k'/, which up until then had been a positional variant of /k/ in other contexts.

(c) 'Morphophonemic genesis'—the recombining of existing distinctive features so as to fill out the pattern of a productive morphophonological alternation. A phoneme is created *de novo*, without there having been a pre-existing (phonetically conditioned) allophone awaiting phonemicization. This process was proposed by Moulton (1961*b*: esp. 243–6), with an example from Swiss German dialects, where the regularly developed umlaut alternation /ɔ/~/ö/ (alongside the regularly developed umlaut alternation /o/ ~ /ö/) was replaced by the alternation /ɔ/~/ɔ̈/, featuring the wholly new phoneme /ɔ̈/. See also Moulton (1967: 1402–5), where he introduced the term and added some analogous Dutch data.

(p.59) III. Via 'foreign loans'—the result of borrowing of words containing the new phoneme (from another dialect or language). An example is the introduction of French loans with initial *v*- into Middle English dialects that lacked initial fricative voicing, which made the sound phonemic. In Modern Engish, /ž/ occurs in French loans such as *prestige*, and has brought within its orbit the medial sound of words such as *vision* (with [-ž-] from [-zj-]), beside *visible* (with [z]) and *visual*, which varies.

Whereas types I and II rely on factors internal to the phonological system, type III relates to external factors and, as a result, does not necessarily present a particularly interesting development structurally (although it may result in hole-filling).

4.5.2 Methodological rider

From a methodological perspective, I suggest one of the ‘analogical’ explanations (type II) should be adduced only if it does not prove possible to account for the phonologization as the result of a merger elsewhere in the system (type I).[19]

4.6 Summary of the process of *a̯*-umlaut

By way of summary, I supply a set of diagrams illustrating my scenario of the phonologization of North-Germanic *a*-umlaut, alongside the development of the vowel system (excluding diphthongs) and illustrated by key forms.

(i) Proto-Germanic (pre-allophone formation)

i:	i	u	u:
e:	e	a (å)	o: (å:)
*/duhtar(-)/	*/hultaz/		
*/duhtariz/	*/hultijaz/		

(ii) North-West Germanic (‘insignificant allophones’)
The shift of $\bar{e}_1$ to $\bar{a}$ (except in final syllables) triggers potential new correlations between short and long vowels and the likelihood of an allophone [o] of short /u/:

	i:	i	u	u:
$\bar{e}_2$ >	e:	e	[o]	o:
		a		
		a:		
*/duhtar(-)/ [dohtar(-)]		*/hultaz/ [holtaz]		
*/duhtariz/		*/hultijaz/		

(p.60) (iii) North Germanic I (‘significant allophones’)
In terms of phoneme inventory, this is the same as North-West Germanic. However, a later generation, aided and abetted by the long-short vowel correlations, ‘perceives’ the [o] allophone of /u/ and levels it within paradigms and between bases and their derivatives.

(The sound change has *ipso facto* ceased to be a 'live rule' by this stage, so the complementary distribution can be disrupted.)

*/duhtar(-)/ [dohtar(-)]	*/hultaz/ [holtaz]
*[doht(a)riz][20] ←	*[holtijaz] ←

(iv) North Germanic II (phonemes)

i:	i	u	u:
e:	e	o	o:
	a		
	a:		

/dohtar(-)/*	/holtaz/*
dohtriR	holtijaR

As both *u* and *o* occur in all positions, the way is open to their extensive shuffling, leading to the observed oddities of their distribution. (This is especially true of *i* and *e*, with the proviso that *e* generally does not occur before following *i* etc., in which case it falls in with /i/.) Redistributed *o* before following *i* etc., instead of reverting to *u*—an option not now open to it, as the allophonic rule is no longer live—persists, later developing the 'front mutation allophone' [ö] in this position (cf. 4.4.1).[21]

Notes:

([1]) The question as to when the attestation starts depends on whether one thinks the inscriptions on the Meldorf fibula (dated to the first half of the first century AD) and the Osterrönfeld pot-sherd (dated to the first century AD) use runic, 'proto-runic', or Latin characters; cf. in the first instance Düwel (2008: 3, 23–4, 178–9). The rough cut-off date of *c*. 500 AD excludes 'transitional' inscriptions that still use the older alphabet but show linguistic innovations; cf. Grønvik (1998: 16–26). Older Runic is remarkably stable linguistically.

([2]) The term '*a*-umlaut' could be justified on the basis that it is generally considered that the loss of this vowel in final syllables triggered phonologization of the change (cf. 4.2.2), but it will turn out that this is in fact not the case. Campbell (1962: §111) refers to 'a tendency to harmonize *e, i*, and *u* to the vowel which follows in the next syllable'. One could dub it the 'North-West Germanic short vowel shuffle', but the term might not catch on. See especially Benediktsson (1970: 100–5). The phonologized change of *a*-umlaut is exclusively a North-West Germanic phenomenon. Any phonemic distinctions between the short vowels /i/ and /e/, and /u/ and /o/ in Bible Gothic (cf. Braune and Heidermanns 2004: §§20, 24) are of independent origin, the result of separate Bible Gothic developments (cf. Bennett 1952, with references). It may be noted that so-called Crimean Gothic is to be classed as a North-West Germanic dialect, as it attests *a*-umlaut; note especially *Schuos* [for *Schnos**]: *Sponsa* 'fiancée' in the list of words not recognized by Busbecq ('cum nostra lingua non satis congruentia'). See Stearns (1978: 151–2), Grønvik (1983: 27).

([3]) I focus on structuralist theory because I consider it the best-fitted to diachronic work, especially the kind of problem under consideration (cf. also Issatschenko 1973: 5). A sociolinguistic approach, by contrast, is not ideally suited for poorly documented or prehistoric stages, although see Ringe (2006*b*) for an inspired example. If they discuss these issues at all, other 'schools' differ largely in using their own terminology and notation. (I allude to some recent work by way of references to Joseph and Janda 2003.) As Jean Rostand observed: 'Le biologiste passe; la grenouille reste'.

([4]) The 'law' is named after its framer, the Russian linguist and orientalist Yevgenij Dmitrievich Polivanov, who promulgated it in an article of 1928 (38 = 1974: 77). See especially Jakobson (1931: section IX); also Hoenigswald (1960: 77–8, 1999: 202–4).

([5]) In my usage, a sound change is 'phonologized' and a specific allophone is 'phonemicized' (achieves phonemic status).

([6]) Runes are transliterated into bold letters, while italics are used for transcriptions, which necessarily involve some degree of interpretation. **R** represents a rune written for etymological *z*, which had acquired some *r*-like features and eventually merged with original *r*. Runic forms are followed in brackets by the name of the monument on which they are found. The standard edition remains Krause and Jankuhn (1966).

([7]) Note also **gudija** ‘priest’, ON *goði* (Nordhuglo), **wurte** ‘(s)he made’, **-kurne** ‘corn’ dat. sg. (Tjurkö), without the conditions for lowering.

([8]) There is no evidence to lead us to believe that 〈**o**〉 is in fact a spelling for /u/ in these examples.

([9]) *Dohtēr* has apparently been relexicalized, which is further evidence of phonologization of *a*-umlaut.

([10]) One of the peculiarities of *a*-umlaut is that, whereas before following *i* and *j* only *i* generally occurs as a reflex, both *e* and *i* are tolerated as reflexes before *a* and *ō* (see the examples in the next paragraph). Compare the treatment of Benediktsson (1967: 183–4). See Hock (1973) for a discussion of the issues and a summary of earlier work.

([11]) Interestingly, some of the early material is in conformity with the later distribution: Gallehus **holtijaR** and **horna** are both from Jutland, Tune **worahto** and **dohtriR** from Norway, and Tjurkö **wurte** and **-kurne** from Sweden. Nordhuglo **gudija** comes from western Norway, however.

([12]) ‘*o*-former som **holtijaR** (der forudsætter **holta* og ikke **hulta*), **worahto** og **dohtriR** der intet som helst kan have at gøre med en formodet *a*-omlyd’. ‘Overgangen af *u* til *o* må være overordentlig gammel i de germanske sprog (sml. gotisk), og den må have været uafhængig af et følgende *a*’.

([13]) As well as Harry Andersen, a vocal opponent of *a*-umlaut was Erik Harding; cf. Harding (1937–49: esp. items 22, 32, 45, 49, 65, 86 (‘det obegripliga [incomprehensible] *a*-omljudet’), 97). Compare also Andersen (1986).

([14]) The following quotation is taken out of context and some clarifying words are added in square brackets in order to help emphasize the desired point. The format is also altered and I silently emend one misprint and change the symbolization to make clear what are (allo)phones (also in square brackets) and what are phonemes (between slashes).

([15]) In the specific context of the creation of a short /o/ phoneme, we may also note that Haas (1978: 246–8) has observed that the quality of a short mid-back vowel in a given phonological system is often tied closely to the quality of the corresponding long mid-back vowel (a point at least anticipated by Moulton, e.g. 1960: 163, 1961b: 236–7, 1970: 23). Similarly, in the case of North-West Germanic, it could be argued that the influence of the long-vowel contrast /iː/ ≠ /eː/ helped preserve that between the short vowels /i/ and /e/ (cf. 4.1.1.2 end).

([16]) 'Perceived allophones', which must be based on a degree of phonetic *dis*-similarity, complement the 'phonetic similarity' criterion for phoneme identification. Others have come to similar conclusions; compare the discussion by Janda in Joseph and Janda (2003: 412–13).

([17]) Compare the discussion of Janda in Joseph and Janda (2003: 415). The idea of the cline and problems of indeterminacy can be illustrated by the oft-discussed situation with [æ] in several varieties of contemporary 'standard English'. The speech-sound occurs in citation forms with three degrees of length: (i) short [æ] is found before voiceless consonants; while both (ii) slightly longer [æˑ] and (iii) fully long [æː] precede voiced segments. Examples are *bat; bade* (also pronounced [beid]); and *bad*. The first two sounds count as short and are automatic positional variants that sound 'the same' to naïve native speakers. The third is perceptibly different. While some words regularly have (iii), e.g. *mad, bad, sad*, there are no rules that predict whether certain other words will have (ii) or (iii). Thus I pronounce *cadge* [æˑ], but *badge* [æː], *mag* [æˑ] (short for *magazine*), but *bag* [æː]. In addition, the lexical incidence of the perceived long allophone varies from speaker to speaker. In my own speech, *brandy* and *shandy* are both pronounced with the long variant, whereas for some speakers they are conveniently distinguished, with *brandy*, a short drink, having variant (ii), while *shandy*, a long drink, has variant (iii); cf. Fudge (1977: 57, 64; to judge by his article, our distribution of (ii) and (iii) differs significantly). There are some potential minimal pairs, for example *candid* [kæˑndid] 'frank' vs. *candied* [kæːndid] 'covered with candy', although, again, I use the long phone (iii) for both (as does Fudge 1977: 58). Granted the existence of a noun *badger* 'someone who puts badges on things' (iii), derived from the verb *to badge*, it would contrast with the animal *badger* (ii) in my speech. There are thus some potential minimal pairs and, on some criteria, (iii) might count as a separate phoneme for some speakers, although not for others. However, the distribution and collocations of the words in question mean that there is no serious functional contrast. Wells (1982: 288) calls it 'marginally contrastive'. Compare the varying views on this issue—which is only raised here to make a general point—of Hoenigswald (1944: 82–3), Bazell (1952: 34; he deems it 'marginal', saying: 'It is neither quite true nor quite false to say that there is an opposition æ/æ: in Standard British pronunciation'), C. A. Ladd (1964: 656), Fudge (1977), Lass (1984: 34–6). A classic example of perceived allophones are the German *Ich*- and *Ach-Laute*; native speakers have no difficulty in hearing the difference between the two sounds, while linguists debate whether they should be considered contrastive and thus different (albeit marginal) phonemes—supposed minimal pairs involve dubious lexemes, diminutive suffixes, or compounds (cf. Issatschenko 1973).

([18]) As Gould (2001: 228) remarks: 'Unpredictable contingency, not lawlike order, rules the pathways of history'.

([19]) In the case of NWGmc *a*-umlaut, a sound change elsewhere may yet be found to have triggered phonologization.

([20]) If we recognize a sound change of PIE **-er(-)* to **-ar(-)* in Proto-Germanic unaccented syllables, the North-West Germanic paradigm of 'daughter' is seen to have lowering environments in the following case-forms: voc. sg. -*ar*, acc. sg. -*ar*, nom. pl. -*ar(z)*, gen. pl. -*rō*. This is a much stronger base—four endings out of nine—from which to generalize *o*-vocalism (cf. the doubts about *a*-umlaut in this form assembled by Andersen 1986: 118–19).

([21]) An earlier version of this paper was presented at the Second International Symposium on Runes and Runic Inscriptions in Sigtuna, Sweden, in September 1985. I would like to thank a number of scholars who have been kind enough to read various drafts or discuss these issues over the years: Michael Benskin, Paul Bibire, N. E. Collinge, Martin Durrell, †Henry M. Hoenigswald, †David McLintock, †William G. Moulton, Lena Peterson.